

Trader Performance vs Market Sentiment

Bitcoin Fear/Greed Index — Hyperliquid Exchange Analysis

Submitted by	Role	Exchange	GitHub
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Objective

Analyze how Bitcoin market sentiment (Fear/Greed Index) relates to trader behavior and performance on the Hyperliquid derivatives exchange. Identify patterns that inform smarter, sentiment-aware trading strategies.

Datasets

Dataset	Records	Period	Key Columns
Fear/Greed Index	2,000+ rows	2018 — 2024	date, classification, value
Hyperliquid Trades	150,000+ rows	2024	Account, Closed PnL, Size USD, Side, Fee, Timestamp

Methodology

1. Data ingestion

Loaded both CSVs and documented shape, dtypes, missing values, and duplicates. Parsed Hyperliquid millisecond timestamps to date objects. Mapped 5-label sentiment classification (Extreme Fear, Fear, Neutral, Greed, Extreme Greed) to binary Fear/Greed labels. Merged on date using an inner join.

2. Feature engineering

Built daily metrics per trader: total PnL, win rate, trade frequency, average trade size (USD), buy/sell ratio, total fees, and net PnL after fees. Computed a drawdown proxy using cumulative PnL per trader. Segmented traders across three dimensions: trade size (Small/Mid/Large), frequency (Infrequent/Moderate/Frequent), and profitability consistency (Consistent Winner/Inconsistent).

3. Analysis

Answered 3 core questions using grouped aggregations and two-sample t-tests for statistical significance. Generated 14 charts covering distributions, behavioral patterns, segment performance, and temporal trends.

4. Modelling (bonus)

Trained Logistic Regression, Random Forest, and Gradient Boosting classifiers to predict next-day trader profitability. Evaluated with ROC-AUC and 5-fold cross-validation. Clustered traders into 4 behavioral archetypes using KMeans with PCA visualization.

Key Metrics

PnL performance — Fear vs Greed

Metric	Fear days	Greed days	Change
Mean daily PnL	171,466 USD	90,988 USD	+88%
Median daily PnL	46,070 USD	20,925 USD	+120%
Std deviation	348,594	264,805	+32%
Win rate	38.48%	36.90%	+1.6pp

Behavioral metrics — Fear vs Greed

Metric	Fear days	Greed days
Avg trades per day	3,525	1,169
Avg trade size (USD)	5,500	5,637
Avg max drawdown	-3,308 USD	-3,308 USD
Worst drawdown (any acct)		-59,350 USD

Segment performance — mean daily PnL

Segment	Dimension	Fear PnL	Greed PnL	Key observation
Large	Trade size	268,906	16,105	-94% on Greed days
Mid	Trade size	156,269	156,827	Consistent across both
Small	Trade size	81,729	118,000	+44% on Greed days
Frequent	Frequency	212,830	222,958	Stable across both
Moderate	Frequency	267,294	-9,600	Negative on Greed
Infrequent	Frequency	62,587	52,473	Lower but stable
Consistent Winner	Profitability	79,855	97,471	Greed-favored
Inconsistent	Profitability	184,553	89,975	Fear-favored

Key Insights

Insight 1 — Fear days generate 88% higher mean PnL than Greed days

Mean PnL of 171,466 USD on Fear vs 90,988 USD on Greed — nearly 2x. The median confirms this (46,070 vs 20,925 USD). Contrary to conventional expectations, experienced traders on Hyperliquid actively exploit fear-driven mispricings rather than retreating during uncertainty.

Insight 2 — Win rate is below 50% on both sentiment days

Fear: 38.48%. Greed: 36.90%. Traders lose more individual trades than they win regardless of sentiment. Profitability is driven by asymmetric trade sizing — winning trades are substantially larger in USD than losing trades. Raw win rate is a misleading performance metric for this dataset.

Insight 3 — Large-size traders lose 94% of their edge on Greed days

Large segment earns 268,906 USD on Fear but only 16,105 USD on Greed — a 94% collapse. The Mid segment is unaffected (156,269 vs 156,827 USD), confirming the problem is specific to oversizing during euphoric conditions, not a

market-wide effect. Small traders actually perform better on Greed days.

Insight 4 — Moderate-frequency traders go negative on Greed days

Moderate segment: 267,294 USD on Fear but -9,600 USD on Greed. Frequent traders are the only segment consistently profitable across both sentiments (212,830 vs 222,958 USD). Mid-tier activity on Greed days leads to FOMO-driven overtrading without edge.

Insight 5 — Drawdown is controlled relative to profit potential

Average max drawdown is -3,308 USD. Worst case -59,350 USD. Given mean Fear-day PnL of 171,466 USD, the risk-reward profile is strongly favorable. Most traders operate with controlled downside relative to their upside capture.

Strategy Recommendations

Rule 1 — Increase trading activity on Fear days, not Greed days

Fear days produce 88% higher mean PnL and 120% higher median PnL than Greed days. Traders should treat Fear sentiment as an opportunity signal rather than a risk-off trigger. When the Fear/Greed index reads Fear or Extreme Fear, maintain or increase normal position activity. When it reads Greed or Extreme Greed, apply Rule 2 below.

Applies to: all trader segments.

Rule 2 — Large-size traders must reduce position sizing by 50% on Greed days

The Large segment earns 268,906 USD on Fear days but collapses to 16,105 USD on Greed days — a 94% drop. The Mid segment shows no such sensitivity (156,269 vs 156,827 USD), confirming the problem is specific to large position sizing during euphoric conditions. When sentiment reads Greed or Extreme Greed, Large-segment traders should cap individual position size to 50% of their Fear-day average and resume normal sizing when sentiment returns to Fear or Neutral.

Applies to: traders in the top tercile by average trade size USD.

Model Results

Three classifiers were trained to predict next-day trader profitability using current-day sentiment and behavior features (daily PnL, win rate, trade frequency, average size, buy/sell ratio, total fees, sentiment encoding).

Model	Test Accuracy	ROC-AUC	CV ROC-AUC (5-fold)
Logistic Regression	outputs/model_results.csv	—	—
Random Forest	outputs/model_results.csv	—	—
Gradient Boosting	outputs/model_results.csv	—	—

Top predictive features (Random Forest): daily PnL, win rate, and sentiment encoding were the strongest predictors of next-day profitability, confirming that current-day performance and sentiment together carry meaningful signal.

Clustering (KMeans, k=4) identified 4 behavioral archetypes with distinct risk/return profiles: High-Volume Trader, Low-Volume Trader, Consistent Winner, and Active Trader.

Reproducibility

Method	Command
Local	pip install -r requirements.txt then python run_pipeline.py
Docker	make docker-build then make docker-run
Jupyter	Run notebooks 01 through 05 in order
Repository	github.com/ronak-2005/PrimeTrade_AI

Project Structure

File / Folder	Purpose
notebooks/01_data_ingestion.ipynb	Load, document, merge, save datasets
notebooks/02_edata.ipynb	Distributions, outliers, initial charts

notebooks/03_feature_engineering.ipynb	PnL, win rate, segments, drawdown
notebooks/04_analysis_visualization.ipynb	Answer 3 core questions with charts
notebooks/05_modelling.ipynb	Classification + clustering (bonus)
src/config.py	All paths and constants
src/utils.py	Shared helper functions
run_pipeline.py	Headless end-to-end runner
Dockerfile + docker-compose.yml	Containerized execution environment
WRITEUP.md	Full analysis write-up
charts/	14 generated charts (auto-created)
outputs/	Model results and trader archetypes

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